

Given $N=1024$, firing rate $f=50\text{ Hz}$ (spikes per sec).

(1)

Mean Aggressive spike rate $= N \times f$

$$R = 1024 \times 50 = 51200 \text{ spikes/sec.}$$

(2)

$$R = 51200 \text{ spikes/sec.}$$

Each AER packet contains 10-bit neuron address
 $= 1024$.

Each packet $= 20$ bits.

$$B = R \times 20 = 1024000 \text{ bits/sec}$$

$$\text{Bit M. bit} = \frac{1024000 \text{ bit/sec}}{10^6}$$

$$= 1.024 \text{ M bits/sec.}$$

③ $B = 1.024 \text{ M bits/sec}$

Compare it to each interface (I²C, SPI, AXI4-Lite)

So $I^2C \leq 3.4 \text{ M bits/s} \rightarrow \text{Sustain } 1.024 \text{ M} \leq 3.4 \text{ M}$

$SPI \leq 50 \text{ M bits/s} \rightarrow 1.024 \text{ M} \leq 50 \text{ M}$

AXI4-Lite $100 \text{ M bits/s} \rightarrow \text{Sustain } 1.024 \frac{\text{M} \cdot \text{bits}}{\text{sec}} \leq \frac{100 \text{ M}}{\text{sec}}$

All 3 - Sustain the mean rate.

The lowest-complexity interface is I²C as the one with the lowest max speed that still exceeds the B. Because if it can handle the bandwidth & it is simplest protocol & 2 wires.

4. Burst Analysis:-

25% of 1024 Neurons fire within 1ms window
= 256

Each packet = 20 bits

Total bits in that burst = $256 \times 20 \text{ bits}$
= 5120 bits.

$$5120 \text{ bits} = 0.00512 \text{ mbits per } 0.001 \text{ sec}$$

$$= 5.12 \text{ mbits/sec}$$

Burst to mean ratio is

$$\text{Peak mbits/sec} \div 1.024 \text{ mbits/sec} = \frac{5.12}{1.024} = 5$$

the burst is 5x times heavier than mean!

T_2C cannot handle the burst ($5.12 \text{ mbits/sec} > 3.4 \text{ mbits/sec}$)

The buffer needs to absorb the extra bits during the burst.

$$\text{Burst peak} = 5.12 \text{ mbits/sec}, \quad T_2C \text{ can drain} = \cancel{3.4 \text{ m}} \text{ mbits/sec}$$

$$\text{excess during } 1 \text{ ms burst} = \frac{(5.12 - 3.4) \text{ mbits/sec}}{0.001}$$

$$= 1.720 \times 10^3$$

$< 1720 \text{ bits}$ of buffering required

5. A conventional sys. Samples 1024 Neurons.

Every 1ms sending 1 bit / neuron / sample.

1ms interval = 1000 samples/sec

$$\text{Frame B.W} = 1024 \text{ Neurons} \times 1 \text{ bit} \times 10^3 \text{ Samples/sec} \\ = 1.024 \text{ Mbit/sec}$$

~~AER to~~

AER to Frame ratio at $f = 50 \text{ Hz}$

$$\frac{\text{AER B.W}}{\text{Frame B.W}} = \frac{1.024}{1.024}$$

$$N \times f \times 20 = N \times 1000 \Rightarrow f = \frac{1000}{20} = 50 \text{ Hz}$$

$f_{\text{crossover}} = 50 \text{ Hz}$

AER is the right choice when the mean firing rate is below 50 Hz.